NITHYA AJAYAN

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EDUCATION

Year	Degree	Institute	CGPA/%
2023 – present	B.Tech (Engineering Physics)	Delhi Technological University (DTU), New Delhi	8.507 CGPA
2023	12th CBSE	Indian School Sohar, Oman	95.4%
2021	10th CBSE	Indian School Sohar, Oman	95.8%

SUMMARY

Engineering Physics undergraduate with experience in astrodynamics, orbit propagation, satellite tracking, and uncertainty analysis. Worked on validating orbital state covariances across LEO, MEO, and GEO regimes using OPM/TLE data, covariance propagation, epoch matching, and Mahalanobis-distance analysis. Contributed to satellite mission research through building transient source localisation algorithms for the Daksha dual-satellite mission complemented by hands-on miniature satellite mission development.

INTERNSHIPS AND WORK EXPERIENCE

Astrodynamics Intern, Digantara, Bangalore *June 2026 - August 2026*

- Validated orbit state covariances from OPM ephemerides across LEO, MEO, and GEO regimes for precise and rapid orbit products.
- Propagated state vectors and covariance matrices using TLEs and OPM reference states, followed by an overlap analysis.
- Computed Mahalanobis distances between overlapping orbital states to evaluate covariance consistency and identify under- or over-dispersed uncertainty estimates.
- Developed a covariance scaling methodology based on statistical discrepancies to improve uncertainty quantification for orbit determination and space situational awareness.

Research Intern, Raman Research Institute, Bangalore *May 2025 – Jul 2025*

- Developed and validated two independent transient localisation algorithms for the Daksha dual-satellite mission, using simulated detector responses across isotropically distributed sky sources.
- Modelled photon count distributions using Band function spectral parameters with realistic detector efficiency and background; recovered source directions with narrow residual spreads closely matching injected values.
- Benchmarked accuracy vs. computational cost trade-offs between the methods.

Embedded Systems Engineer Intern, India Space Lab, Delhi *Sep 2024 – Dec 2025*

- Architected and integrated the embedded systems of three miniature satellites, including sensor suites, power subsystems, and a recovery system.
- Established and validated long-range radio telemetry links for real-time onboard data downlink; conducted rigorous end-to-end testing of the acquisition and transmission pipeline.
- Conducted multiple workshops for college students from various universities about the basics of cansat configuration and assembly.

Embedded Systems Engineer Intern, Astroverse Pvt. Ltd., Nainital *Feb 2024 – Mar 2024*

- Built a fully functional miniature satellite prototype; secured long-range radio communication between satellite and ground station for live telemetry.
- Conducted rigorous pre-launch testing of the telemetry pipeline under simulated operational conditions.

PROJECTS

Exoplanet Vetting Using Neural Networks

- Built MLP Classifier and XGBoost models to distinguish confirmed exoplanets from false positives across Kepler, K2, and TESS mission datasets; tested multiple architectures and achieved 92% classification accuracy.
- Performed feature engineering on photometric light curves and transit parameters using `astropy` and `scikit-learn`.

Solar Flare Classification

- Developed a CNN-hybrid model to classify solar flares (A/B/C/M/X classes) using HMI solar magnetogram images from SDO.
- Applied transfer learning and data augmentation to address class imbalance across flare severity levels.

POSITIONS OF RESPONSIBILITY

Captain, Team CanSat DTU

- Leading a 20+ member interdisciplinary team for model satellite missions; overseeing avionics, mechatronics, software, corporate, and media.

General Secretary, Cosmology Club DTU

- Organised sessions on cosmology and observational astronomy; coordinated talks with research scientists and astronomers.

Vice President, Deltech Engineering Physics Technical Hub

- Designed and conducted technical events for inter-college fests; facilitated group learning sessions on physics topics.

TECHNICAL SKILLS

- **Astrodynamics:** Orbit propagation, OPM, TLEs, covariance propagation, orbit determination, uncertainty quantification, Mahalanobis distance, conjunction assessment, SSA
- **Scientific Computing:** Python, NumPy, SciPy, Pandas, Matplotlib, Jupyter, MATLAB
- **Astronomy & ML:** Astropy, healpy, scikit-learn, PyTorch, TensorFlow, XGBoost
- **Satellite Systems:** CubeSat/CanSat integration, telemetry, sensor interfacing, Arduino, embedded C, SolidWorks, Ansys HFSS
- **Coursework:** Cosmology & Astrophysics, Probability & Statistics, Stochastic Processes, Communication Systems, AIML